

PAX placement: before and after

COMPARISON

Before: PAX closer to the interferometer. After: PAX displaced and placed on a heavier base. Placement descriptions are supplied by the operator; distance and base mass were not recorded.

SOURCE EXPERIMENTS

Before: 203850_pax-vibration_pax-vibration

After: 235148_pax-vibration_pax-vibration

Both completed; this is offline analysis of the saved raw measurements.

MAIN FINDING

Fast PD noise is substantially lower after relocation. The normalized 5–1000 Hz motor-on/off variance ratio falls from 8.04 to 1.63. Excess spectral power near 228 and 238 Hz is strongly suppressed.

POLARIZATION FINDING

The mean Stokes directions differ by 113.1 degrees on the Poincare sphere. Angular RMS spread rises from 5.99 to 13.85 degrees; mean DoP falls from 94.3% to 87.3%. Polarization characteristics were not preserved.

INTERPRETATION

Consistent with reduced mechanical coupling, but not a controlled isolation of that cause: position and base changed together, and the optical operating point changed. These are two sequential runs, not independent replicated placement trials.

MATCHED ACQUISITION

IN2; zero static outputs; 30 s requested per motor state; 5 s settling after motor stop; 15,258.8 samples/s; 16,384 samples per capture; 830 nm PAX setting. Motor off means rotation stopped, not electrical power removed.

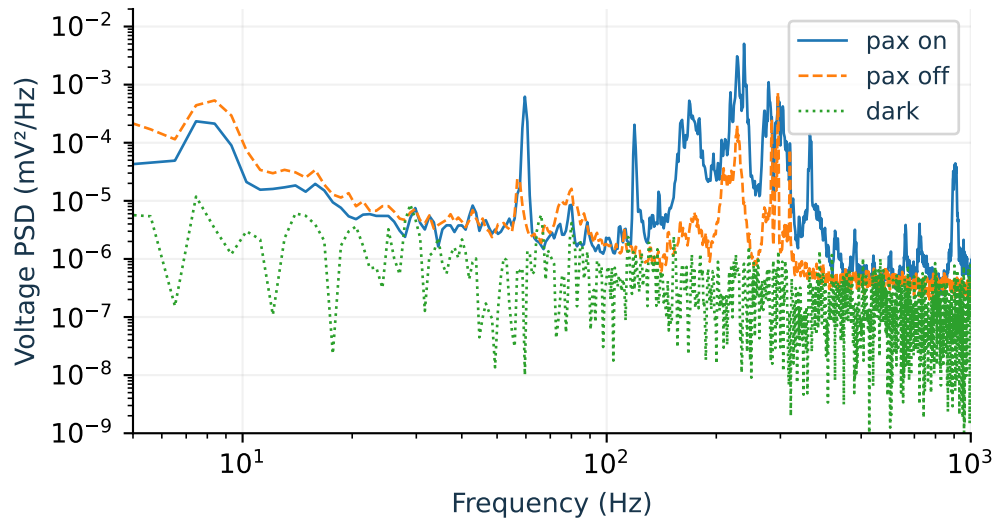
Quantitative comparison

Metric	Before: near	After: displaced + base
PD band RMS, motor on	234.2 μ V	52.8 μ V
PD band RMS, motor off	79.5 μ V	41.5 μ V
PD relative band RMS, motor on	2.61%	0.432%
PD relative band RMS, motor off	0.921%	0.339%
Normalized band variance: on / off	8.04	1.63
PD light signal, motor on	8.97 mV	12.22 mV
Mean PAX optical power	589.8 μ W	294.8 μ W
Mean DoP	94.3%	87.3%
Stokes angular RMS spread	5.99 deg	13.85 deg
Successive-reading angular RMS step	5.01 deg	3.86 deg
PAX power relative RMS	6.97%	7.57%
PD captures, on / off	26 / 28	25 / 27
PAX readings / approximate rate	208 / 6.77 Hz	200 / 6.60 Hz

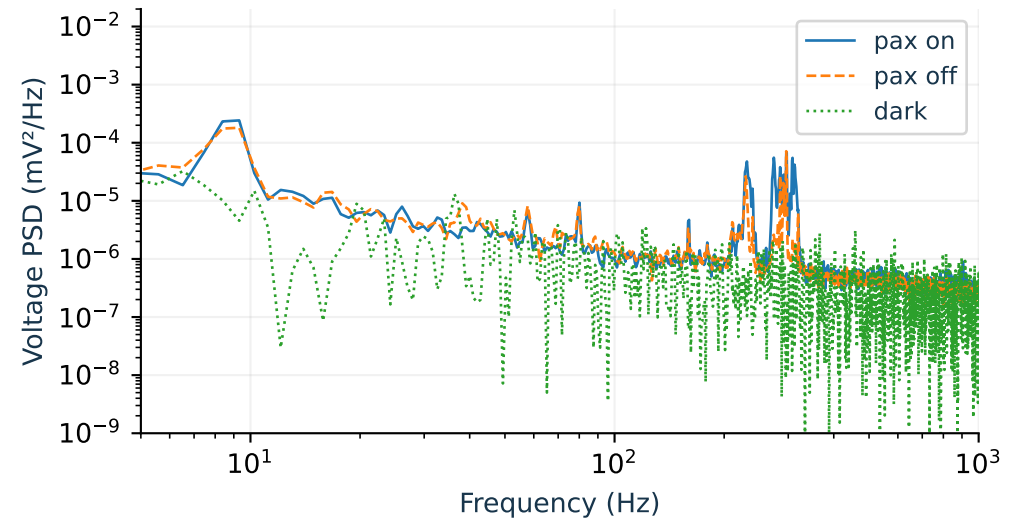
PD metrics use the 5–1000 Hz band. PAX metrics use its slow, motor-on readings only.
 Relative PD power uses the measured dark-corrected mean; voltage noise is not fringe-contrast variance.

PD spectra: suppression of the original peaks

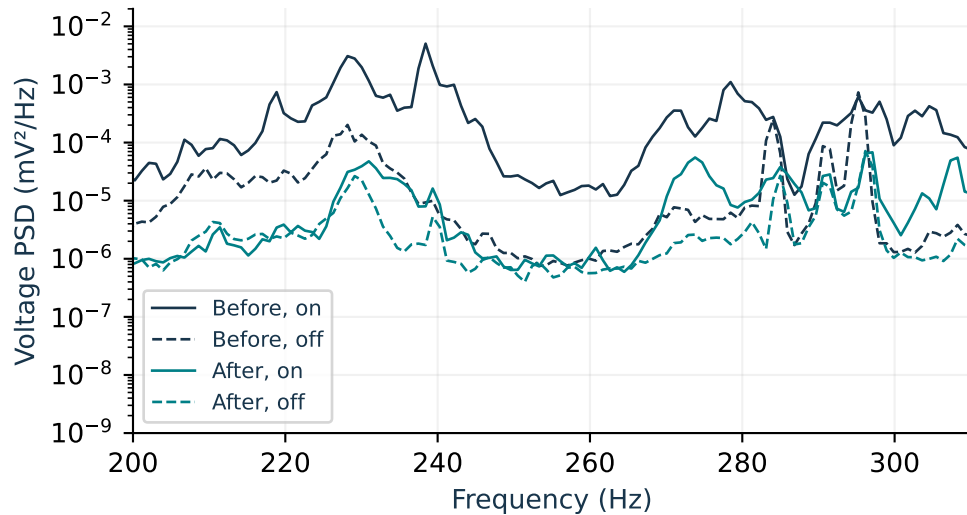
Before: near interferometer



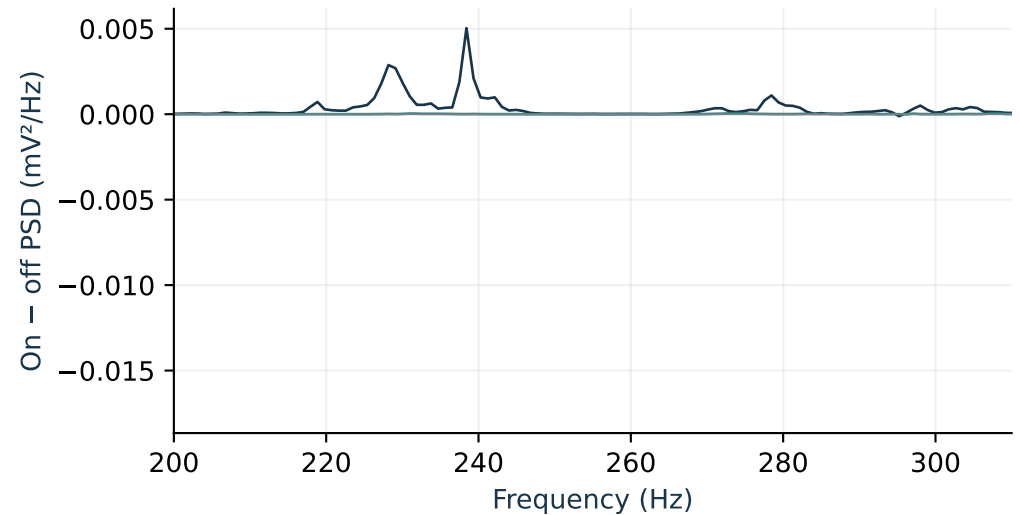
After: displaced, heavier base



Same axes: 228 / 238 Hz region

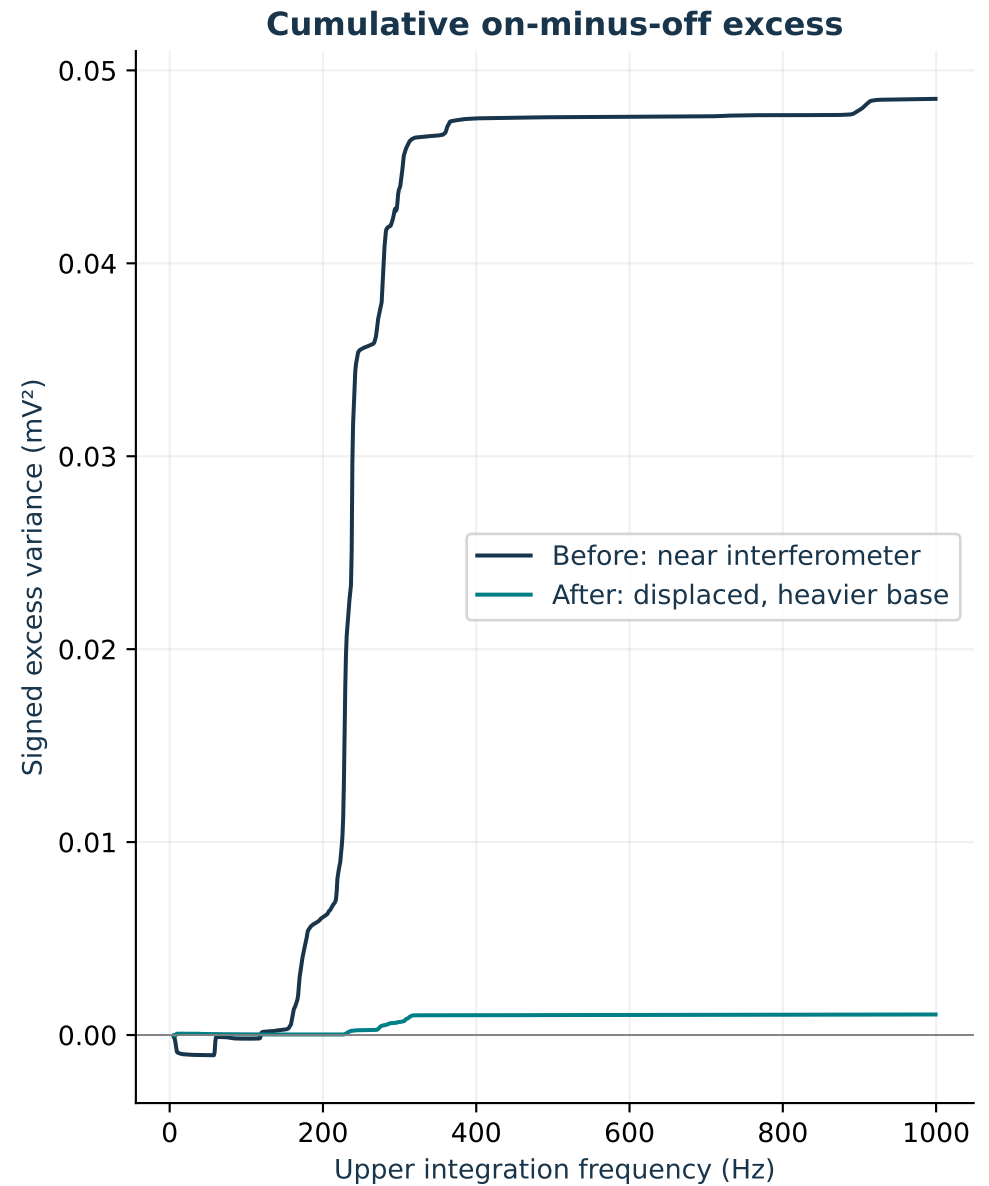
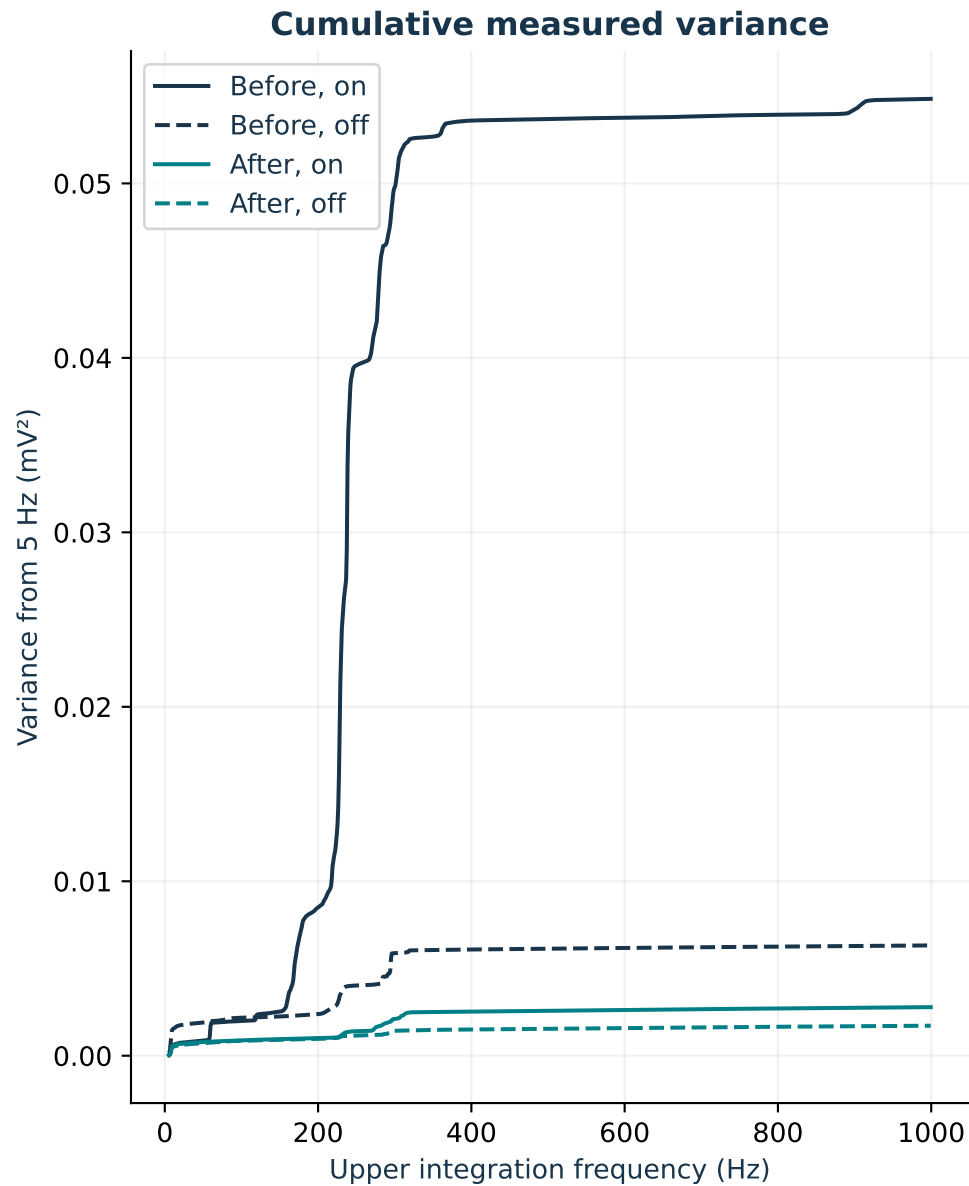


Signed motor-associated excess



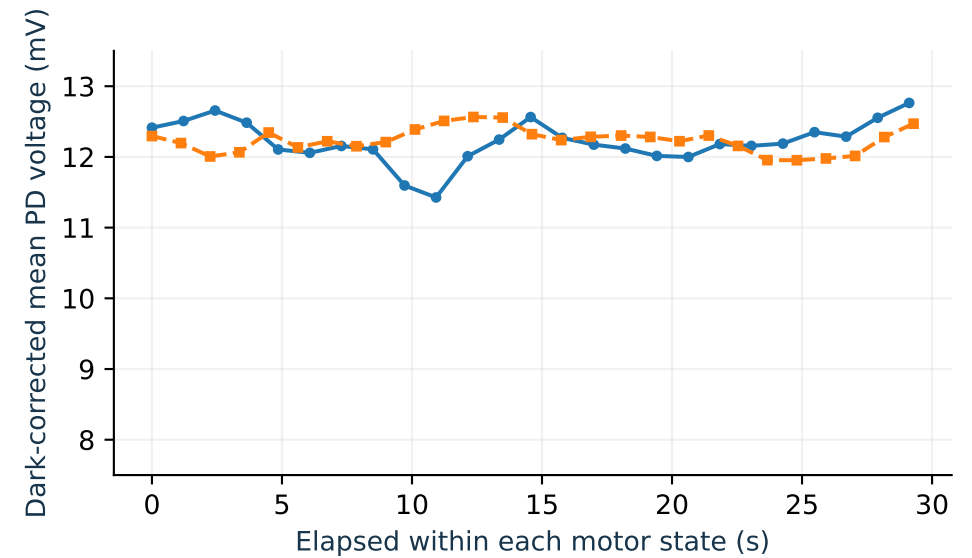
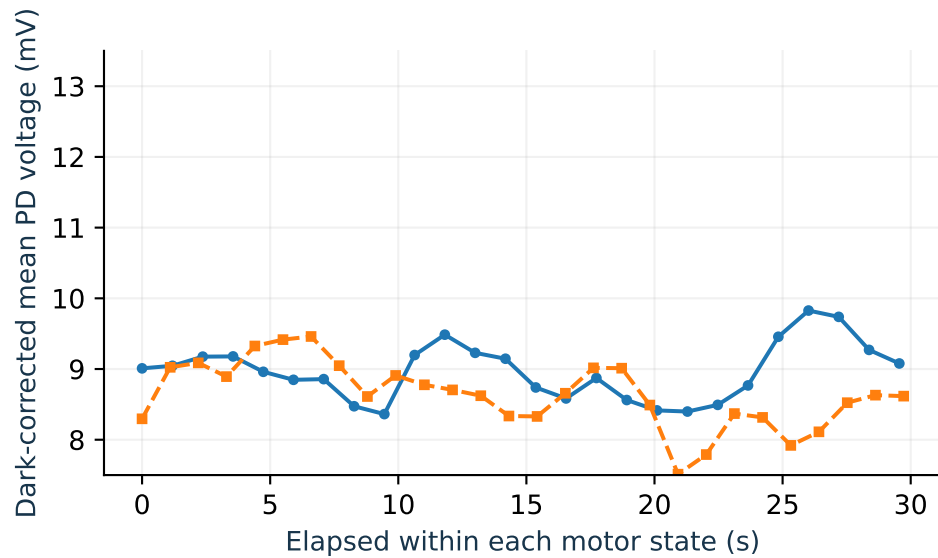
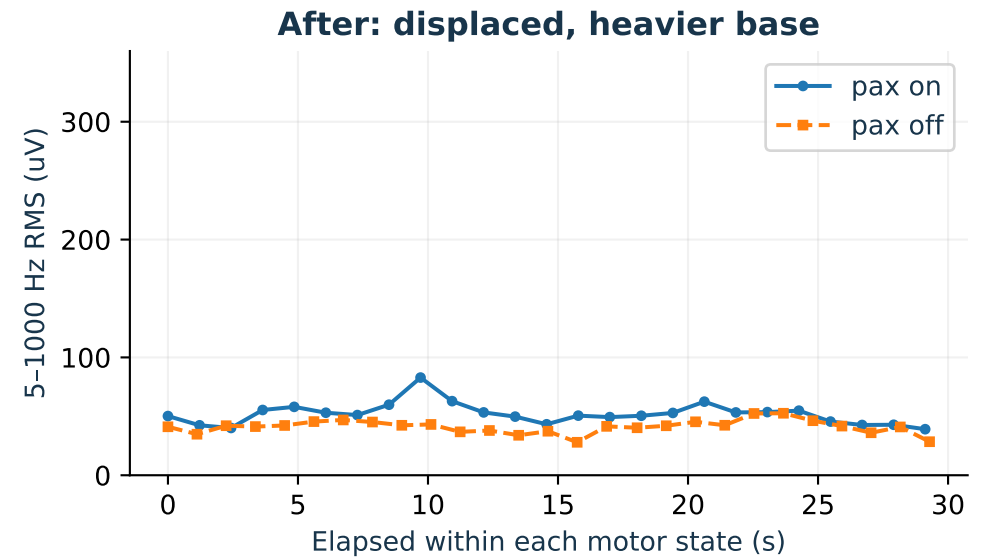
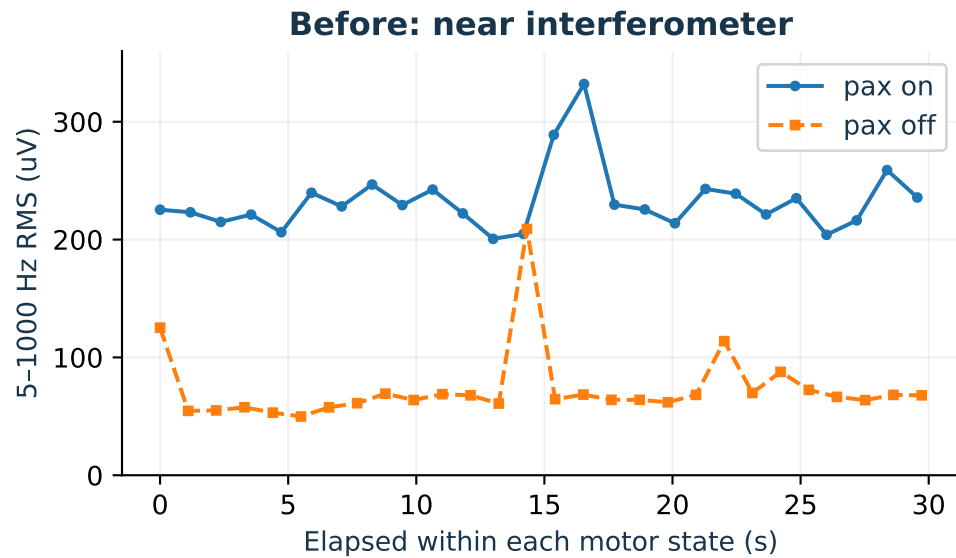
Identical Hann periodogram method and frequency grid (~ 0.93 Hz bins); same vertical scales in the log plots. Dark traces show the electronics baseline. Frequency peaks alone do not identify mechanical causation.

Where the noise power accumulates



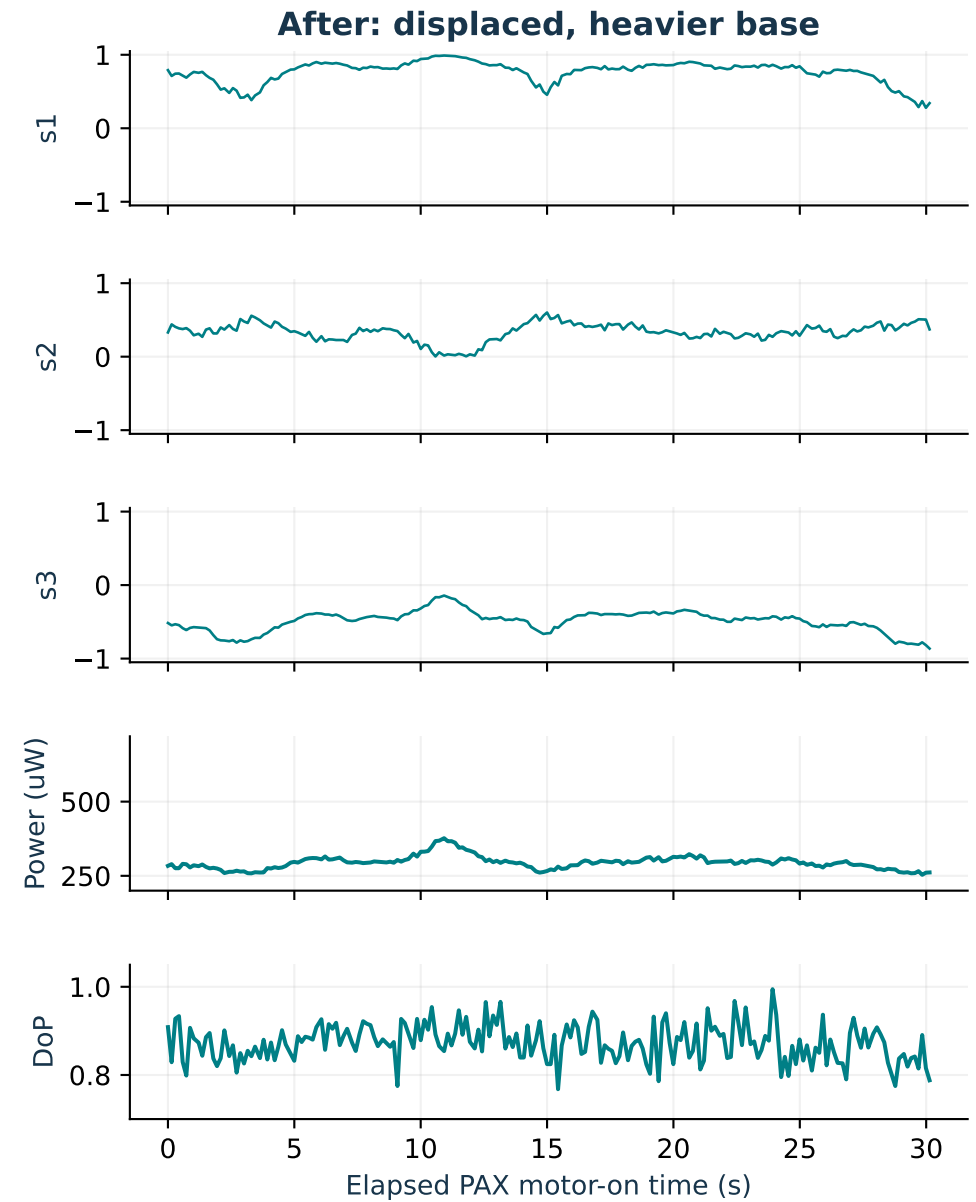
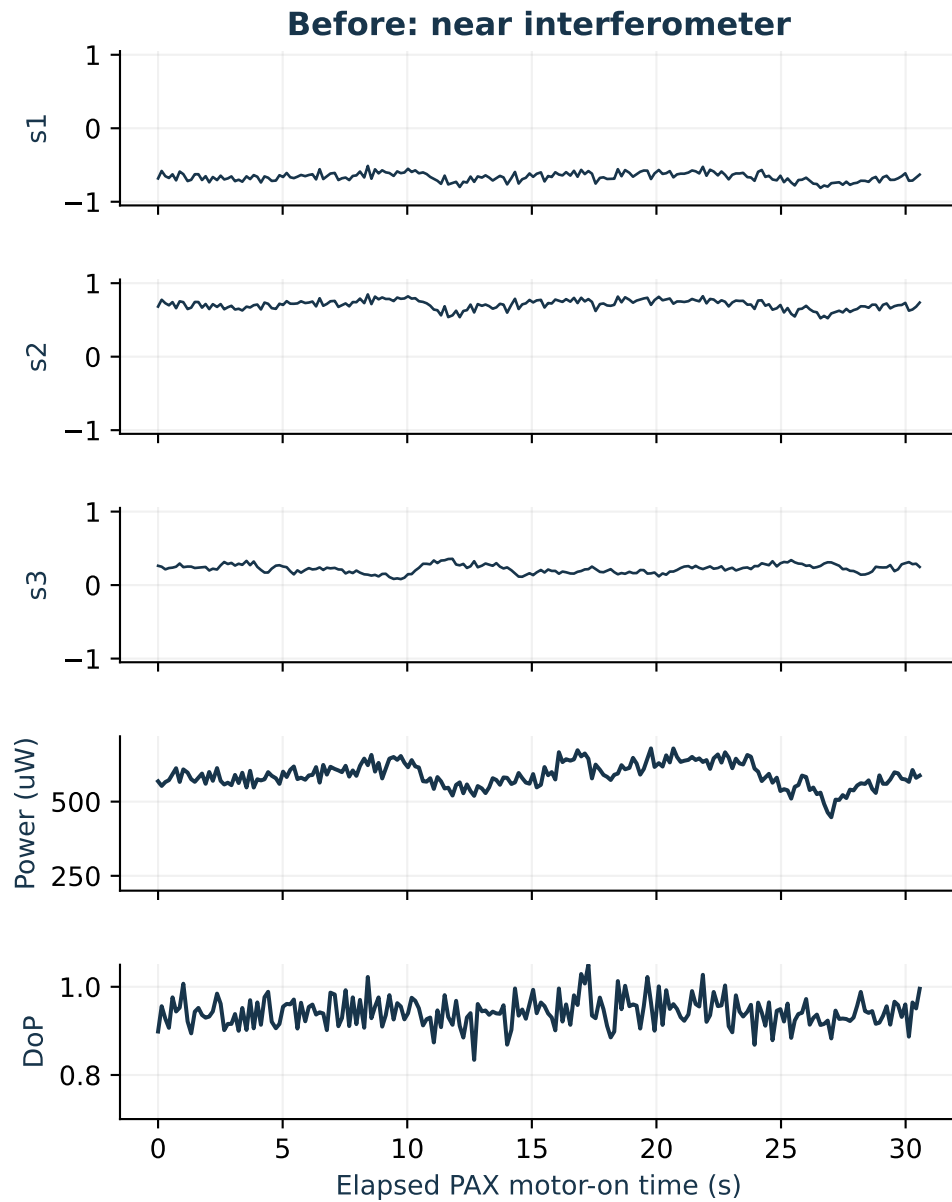
Integration starts at 5 Hz. Sharp rises identify frequency regions contributing most of the noise.
Negative contributions are retained; excess power is not clipped or claimed to be purely mechanical.

PD trace width and light level across captures



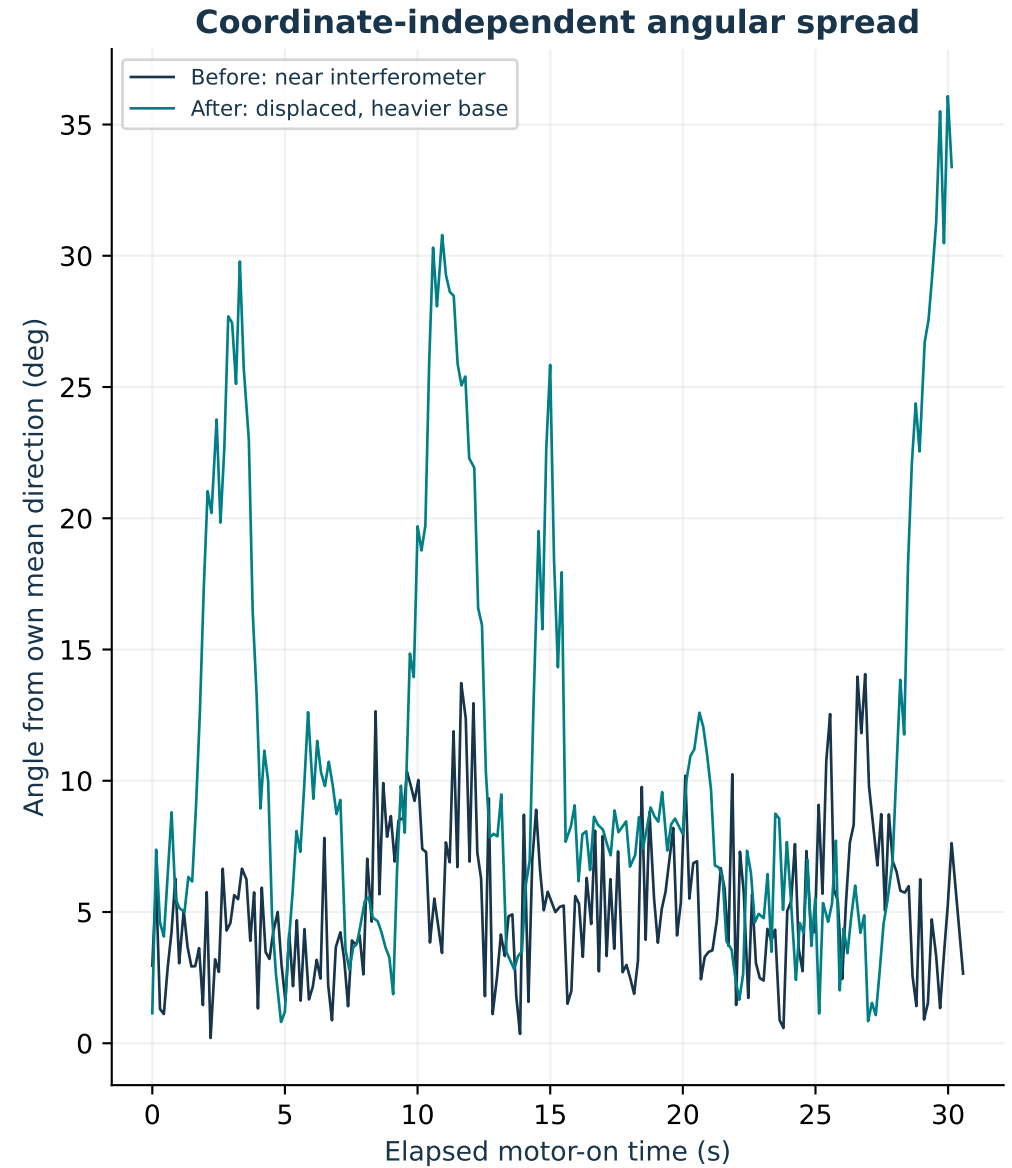
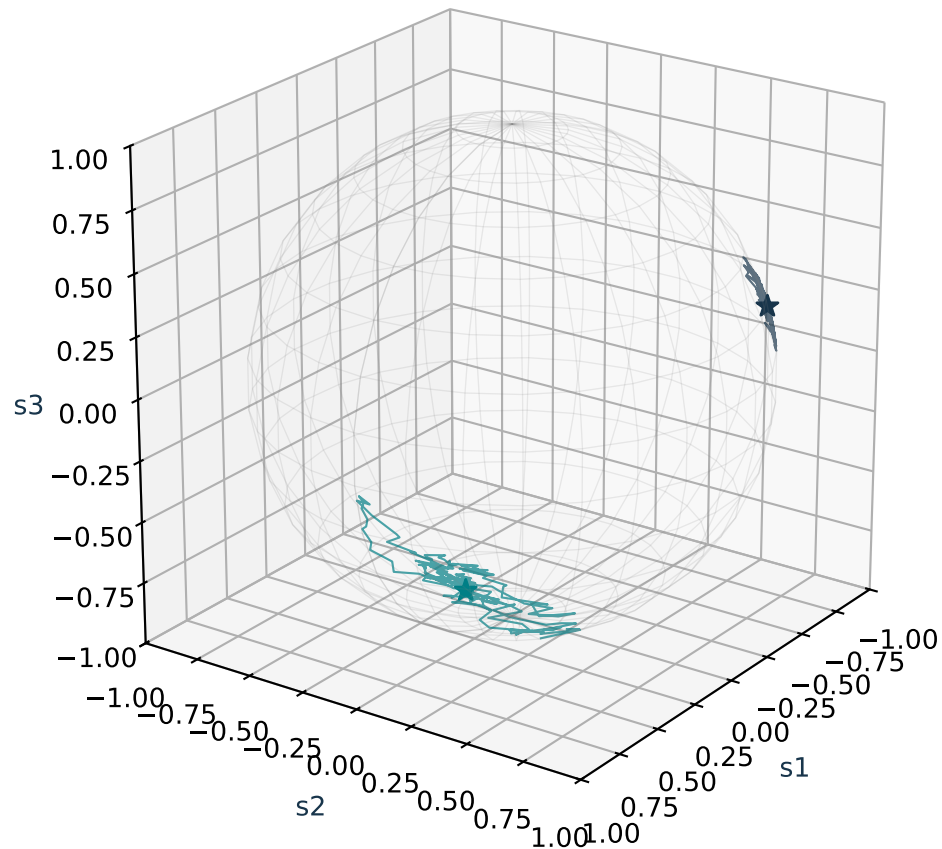
Each point is one ~ 1.074 s FPGA capture, not an independent placement trial. Motor windows are sequential. The PD light signal is larger after relocation; the lower voltage noise is not explained by a dimmer PD signal.

PAX trajectories: polarization, power and DoP



Normalized Stokes direction is plotted separately from DoP. Identical axis limits enable direct comparison.
PAX samples at $\sim 6.6\text{--}6.8$ Hz: these trajectories cannot resolve the PD peaks near 228 and 238 Hz.

Polarization geometry and angular fluctuations



Mean directions are 113.1° apart; stars mark normalized mean directions, not polarization magnitude.
Angular RMS: 5.99° before, 13.85° after. Larger slow excursions can coexist with suppressed fast PD noise.

Methods, provenance and interpretation

PD CALCULATION

Signed voltages are retained. Each capture is mean-subtracted, Hann-windowed and converted to a one-sided periodogram. Periodograms are averaged within each motor state. Integrate 5–1000 Hz; RMS is the square root of variance. Normalize by the square of each state's dark-corrected mean voltage. No dark-noise PSD subtraction is applied.

PEAK COMPARISON

Integrate signed on-minus-off PSD over ± 3 Hz around 228.174, 238.419 and 278.465 Hz. The first two peak-band excess powers decrease by about 144x and 192x. These are descriptive comparisons, not confidence bounds or estimated mechanical transfer functions.

STOKES CALCULATION

s1–s3 are the suite's normalized Stokes direction; DoP is separate. Mean directions are normalized vector averages. Angular spread is RMS geodesic distance to each run's own mean direction. Successive-step RMS uses adjacent readings at similar, but not identical, update rates. Slow drift and measurement noise are both included.

LIMITS ON ATTRIBUTION

Position and supporting base changed together. Optical power, mean polarization and DoP also changed; a different optical operating point can change sensitivity to phase and polarization motion. One run per placement supplies no placement-level replication. Do not describe this as contrast variance or proof of a mechanical frequency.

NEXT DISCRIMINATING MEASUREMENT

Repeat alternating placements or isolation states while preserving optical alignment, polarization operating point, PD input and gain. Record a vibration sensor synchronously if identifying mechanical coupling is the objective.

VERIFICATION AND REPRODUCIBILITY

Raw CSV values and setup metadata were read directly. Recomputed PD statistics and PAX summary metrics match the saved run summaries to $1e-12$ relative tolerance. Source hashes and software versions are in provenance.json. analyze.py regenerates this report without instrument access; source experiments are not

Methods, provenance and interpretation

modified.